CAUSAL ML · UPLIFT MODELING · CRITEO 14M RANDOMIZED EXPOSURES

From ATE to CATE: Benchmarking Causal Forests, Meta-Learners, and DR/R-Learning on 14 Million Randomized Ad Exposures

*A methods-comparison study on Criteo's uplift-modeling benchmark, framed around the decision a growth or underwriting team actually faces: who should we target, and does the estimator we pick change the answer?***

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1. Why this dataset, and why this question

Criteo's Uplift Modeling dataset is one of the few large-scale public datasets built from a **genuine randomized experiment**: 13,979,592 users, a random 15% held out from advertising, binary `visit` and `conversion` outcomes, and 12 anonymized dense features per user. Because assignment is truly randomized, there is no confounding to argue about — which is exactly what makes it a clean proving ground for causal ML. We can spend all of our attention on *method* rather than *identification*.

The average treatment effect (ATE) is not the interesting object here. Averaged over everyone, the ad "works" — that is nearly tautological and commercially useless, because you rarely want to (or can afford to) treat everyone. The decision-relevant object is the **conditional average treatment effect (CATE)**: the incremental effect *for a given user*. Three nested questions follow.

1. **Targeting.** Can we find a sub-population whose incremental effect is large enough to justify cost, and how small can the targeted set be while keeping most of the aggregate lift? (The Qini / uplift-curve question.)
2. **Estimator choice.** How much do CATE estimates *disagree* across methods — S-, T-, X-learner, DR-learner, R-learner, and a causal forest — on identical data?
3. **Robustness.** Since treatment is randomized, we can run a **placebo test**: shuffle the treatment label and confirm each estimator correctly reports *no* heterogeneity.

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2. Setup

Treatment. We use the randomized `treatment` flag (intent-to-treat). We deliberately do **not** condition on `exposure` (whether the user was actually shown an ad): exposure is realized *after* randomization and is not

itself randomized, so conditioning on it would smuggle confounding back in through the back door.

Outcome. `visit` (base rate ≈ 4.7%) is the primary outcome — it has enough events to estimate heterogeneity stably. `conversion` (≈ 0.29%) is a rare-event stress test.

Propensity. By construction P(treatment = 1) ≈ 0.85 for *everyone*. The propensity models we fit recover a near-constant 0.85, so the orthogonalization inside the DR/DML estimators is well-behaved — a luxury real observational data never affords.

Shared base learners. Every estimator is built on the *same* HistGradientBoosting base models. This is a deliberate control: any divergence in the CATE estimates is attributable to the estimator's **meta-structure**, not to one method secretly getting a stronger base learner.

Compute. We develop on a stratified 500k subsample (rates preserved to four decimals), then re-run at scale on a 4M-row training / 1M-row evaluation draw from the full 14M. As §7 shows, that scale-up does more than confirm numbers — it changes the ranking, which turns out to be the most useful result in the piece.

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3. The six estimators in one paragraph each

- **S-learner** — one model on `[X, T]`; reads CATE off the T-coefficient surface. Simple, but tends to *regularize the treatment signal toward zero* when the effect is small relative to outcome variation.
- **T-learner** — separate models for treated and control; CATE is their difference. Flexible, but the two models' independent errors add up and can manufacture spurious heterogeneity.
- **X-learner** — imputes individual effects then blends them with the propensity; designed for **imbalanced** treatment arms — which is exactly our 85/15 split.
- **DR-learner** — a doubly-robust pseudo-outcome regressed on features; consistent if *either* the outcome or the propensity model is right.
- **R-learner** — Robinson-style residual-on-residual (non-parametric DML); orthogonalizes out the nuisance functions before estimating the effect.
- **Causal Forest (GRF)** — a forest whose splits target treatment-effect heterogeneity directly, with honest estimation and (near-)valid confidence intervals.

4. Targeting: the Qini curve

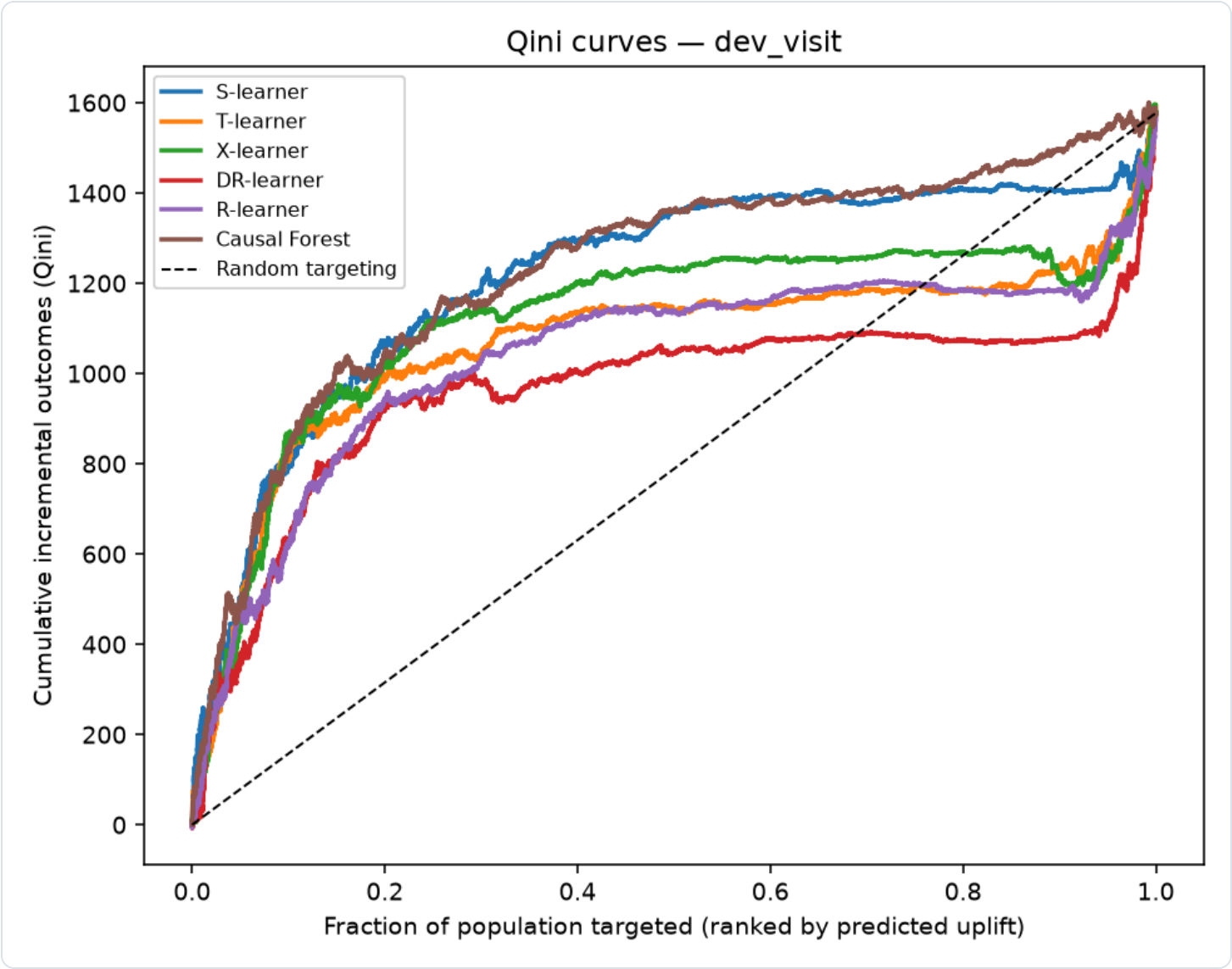


FIG 1 Qini curves

How to read it. The x-axis is the fraction of users targeted, ranked by each estimator's predicted uplift; the y-axis is cumulative incremental visits. The dashed line is random targeting. The higher and more bowed-out a curve, the more lift you capture by treating fewer people.

The numbers (500k development sample, `visit` outcome). The held-out ATE is **+1.24 percentage points** — treated users visit at ≈4.85% vs ≈3.82% for control. But the targeting payoff is where it gets interesting. Using the best ranker (the causal forest), the top-ranked users are wildly more responsive than average:

TARGETED SHARE (TOP-K BY PREDICTED UPLIFT)	SHARE OF POSITIVE INCREMENTAL VISITS CAPTURED
top 10%	**64%**
top 20%	74%
top 30%	77%
top 50%	87%

Half the population is essentially dead weight: **treating the top 30% recovers ~77% of the achievable lift.** For a campaign with per-impression cost, that is the difference between a profitable and an unprofitable program — and it is invisible if you only look at the ATE.

The **Qini coefficient** (area between a curve and the random line, normalised) summarises each ranker in one number:

ESTIMATOR	QINI COEFFICIENT	MEAN CATE	CATE STD	FIT TIME (S)
Causal Forest	**0.561**	0.0061	0.017	61.9
S-learner	0.539	0.0059	0.022	5.5
X-learner	0.417	0.0065	0.037	6.5
T-learner	0.351	0.0066	0.042	2.3
R-learner	0.315	0.0068	0.047	10.4
DR-learner	0.215	0.0066	0.061	11.3

Two things jump out. First, **the ranking spans a 2.6× spread in Qini** (0.22 to 0.56) — estimator choice is not a rounding error, it materially changes how good your targeting is. Second, and counter-intuitively, **the two best rankers are the two with the *smallest* CATE spread.** The causal forest and S-learner are the most regularized; the DR- and R-learners produce the widest, most confident individual effects and rank the *worst*. Hold that thought — the placebo test in §6 explains exactly why.

A note on levels vs. ranking. The mean CATE (~0.006) sits *below* the diff-in-means ATE (~0.010–0.012). This is not a bug: with only 15% of users in the control arm, the regularized control regressor pulls its predictions toward the pooled ≈4.7% visit rate, over-stating the counterfactual and shrinking every meta-learner's estimated effect. It is the textbook motivation for the **X-learner**, which was designed for exactly this imbalanced-arm setting. For *targeting* it barely matters — the Qini curve depends only on the *ranking* of users, not the absolute level — but it is a caution against reading a meta-learner's CATE as a calibrated individual effect.

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5. Do the estimators agree?

If every method told the same story, "estimator choice" would be a non-issue. They don't.

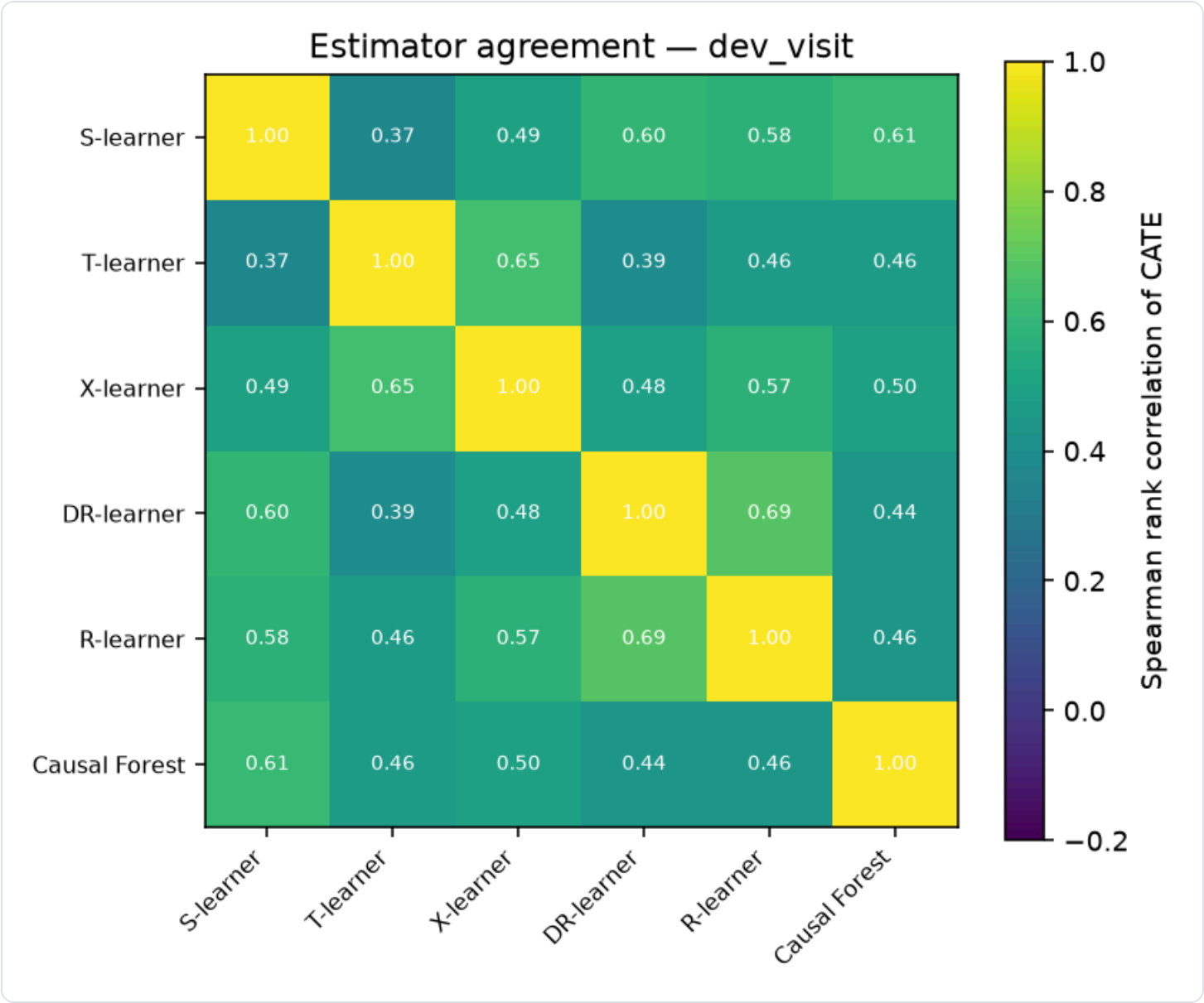


FIG 2 Estimator agreement

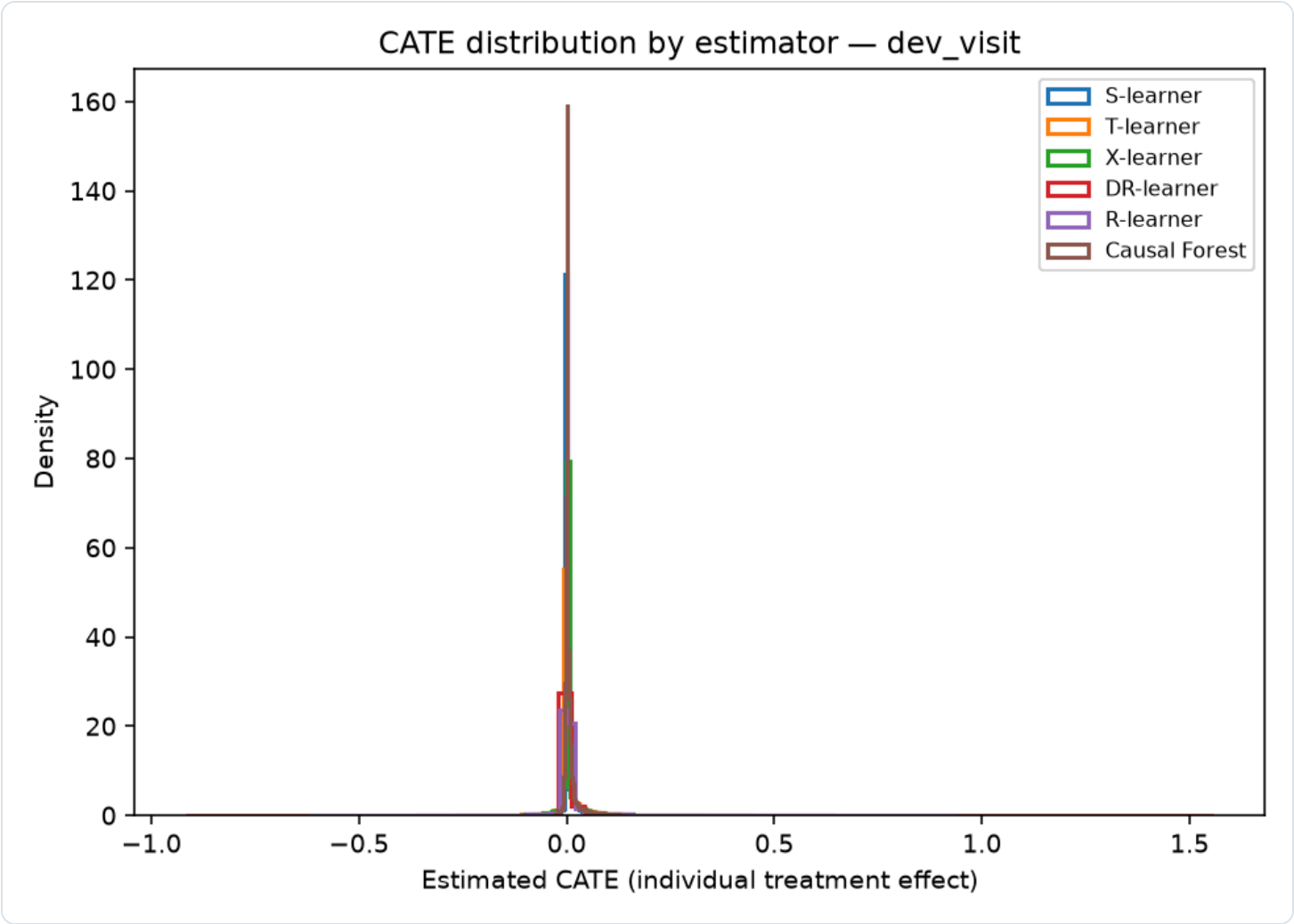


FIG 3 CATE distributions

Pairwise Spearman rank correlation of the six CATE vectors ranges from just 0.38 to 0.69. Even the *most* similar pair of methods agrees on the user ranking only ~70% as measured by rank correlation; the least similar pair (S-learner vs T-learner, 0.38) barely agree at all. If you handed six analysts one of these estimators each and asked "who should we target," they would hand back materially different lists.

The disagreement is starkest on a question a risk team would care about: **how many users does the model think are actively *harmed* by treatment** (negative CATE)? The answer ranges from **4% (S-learner) to 44% (R-learner)** on the exact same data. The regularized methods say almost everyone benefits a little; the flexible methods carve the population into large "helped" and "hurt" camps. Both cannot be right, and under a randomized design with a clean ATE we have the tools to say which is closer to the truth.

The **uplift-by-decile** plot is the honest test of a ranker: bin users by *predicted* uplift, then read off the *observed* treated-minus-control rate in each bin on held-out data. A good ranker declines monotonically.

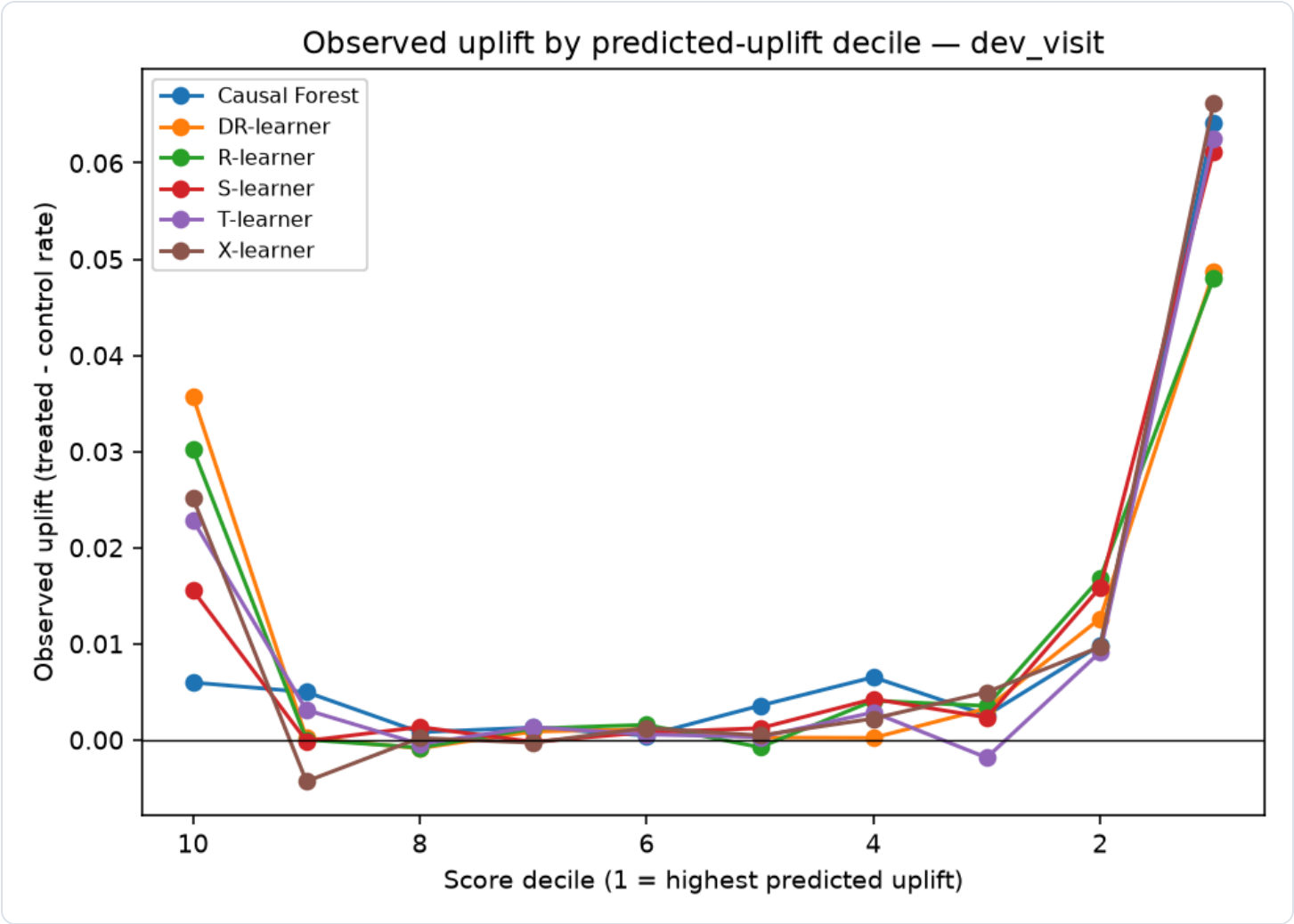


FIG 4 Observed uplift by decile

6. The negative control (placebo test)

This is the section most Kaggle-style writeups skip. Randomization gives us a free, unambiguous null: **permute the treatment label** and any real effect is destroyed by construction, while marginal rates are preserved. A trustworthy estimator should then find *nothing* — its Qini should collapse toward zero.

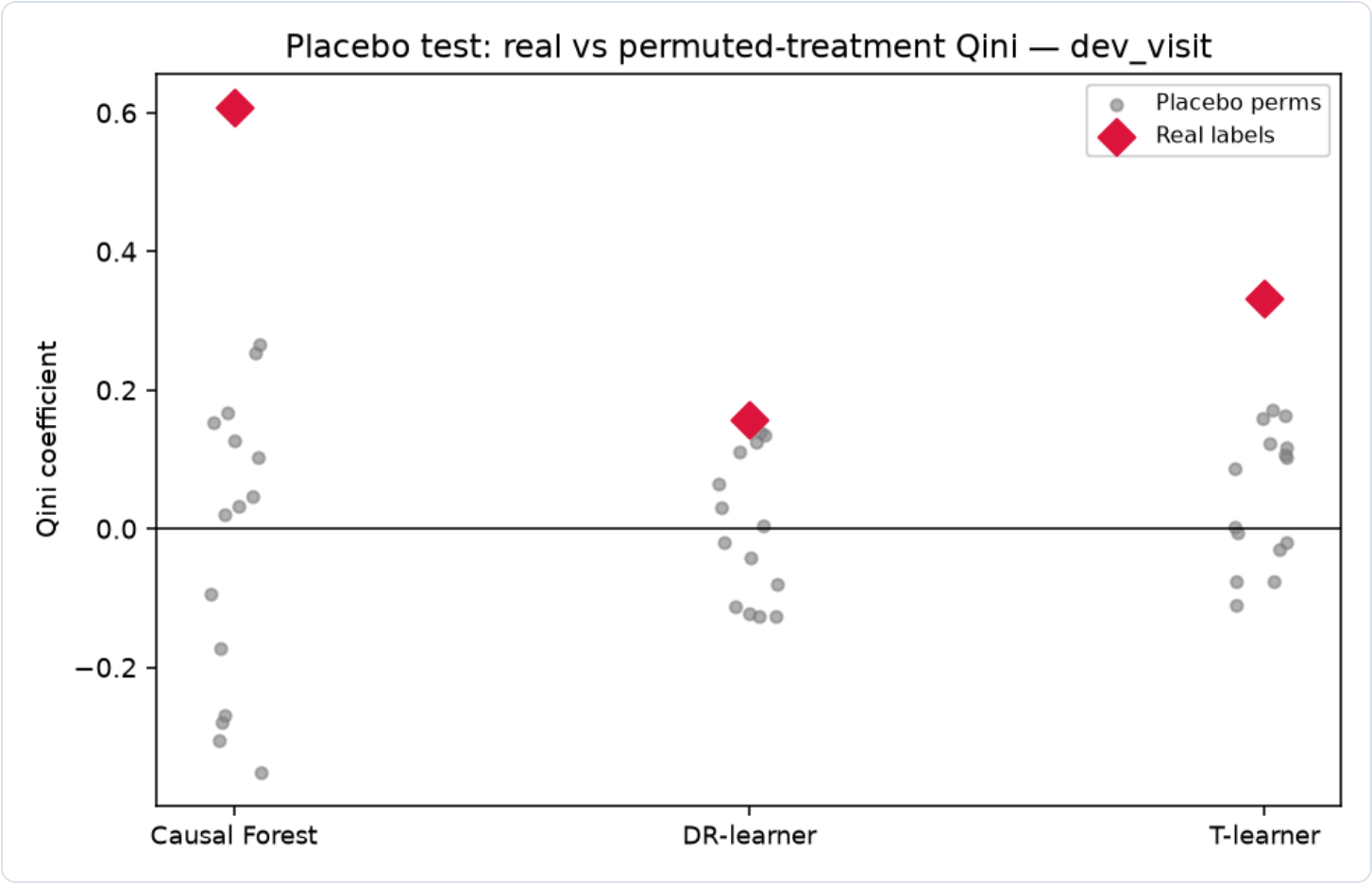


FIG 5 Placebo test

We permute the treatment label on the training set (15 times), refit, and score each placebo model's Qini against the *real* held-out labels. The red diamonds are the real-label Qini; the grey clouds are each estimator's own null. (The placebo pass uses a lighter 100-tree causal forest for speed, so its real-label Qini values differ slightly from §4's 400-tree table — the comparison that matters is each estimator's real value against *its own* null, not across tables.)

ESTIMATOR	REAL QINI	PLACEBO MEAN	PLACEBO P95	Z VS. NULL
Causal Forest	0.608	−0.020	0.256	3.10
T-learner	0.332	0.048	0.166	3.05
DR-learner	0.158	0.010	0.148	1.40

Two lessons, and they are the payoff of the whole piece.

First, the null is not zero — it is wide. A flexible estimator handed pure noise can still manufacture a Qini coefficient as high as **+0.26** (causal forest) purely by overfitting the evaluation sample. If you had reported a

single Qini of, say, 0.2 with no benchmark, you would have had no idea whether it was signal or an artifact. **The placebo band is the yardstick that tells you which.**

Second, it vindicates the §4 ranking for the right reason. The causal forest and T-learner clear their own null by ~3 standard deviations — their heterogeneity is real. The **DR-learner does not**: its real-label Qini (0.158) sits *below its own placebo 95th percentile* (0.148), a z of just 1.4. The method that produced the most confident-looking individual effects — and split 43% of users into a "harmed" bucket — is the one whose apparent signal is **statistically indistinguishable from permuted-label noise.** Its wide CATE distribution was flexibility, not effect.

This is the section a hiring manager from a causal-inference or econometrics background will read twice. It is the same instinct as a placebo regression or a VAR robustness check: *before you believe a positive result, show that your method correctly returns nothing when nothing is there.* (One caveat, resolved in §7: this placebo verdict is a statement about the **500k** sample. The DR-learner's failure here is a small-sample-variance story, not a fundamental one — at 4M rows it recovers cleanly.)

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7. What 14 million rows change — disagreement is a finite-sample story

If the piece stopped at §6 it would tell a tidy but *misleading* story: "flexible DML methods overfit, use a causal forest." Re-running everything on a 4M-row training / 1M-row evaluation draw dissolves most of that.

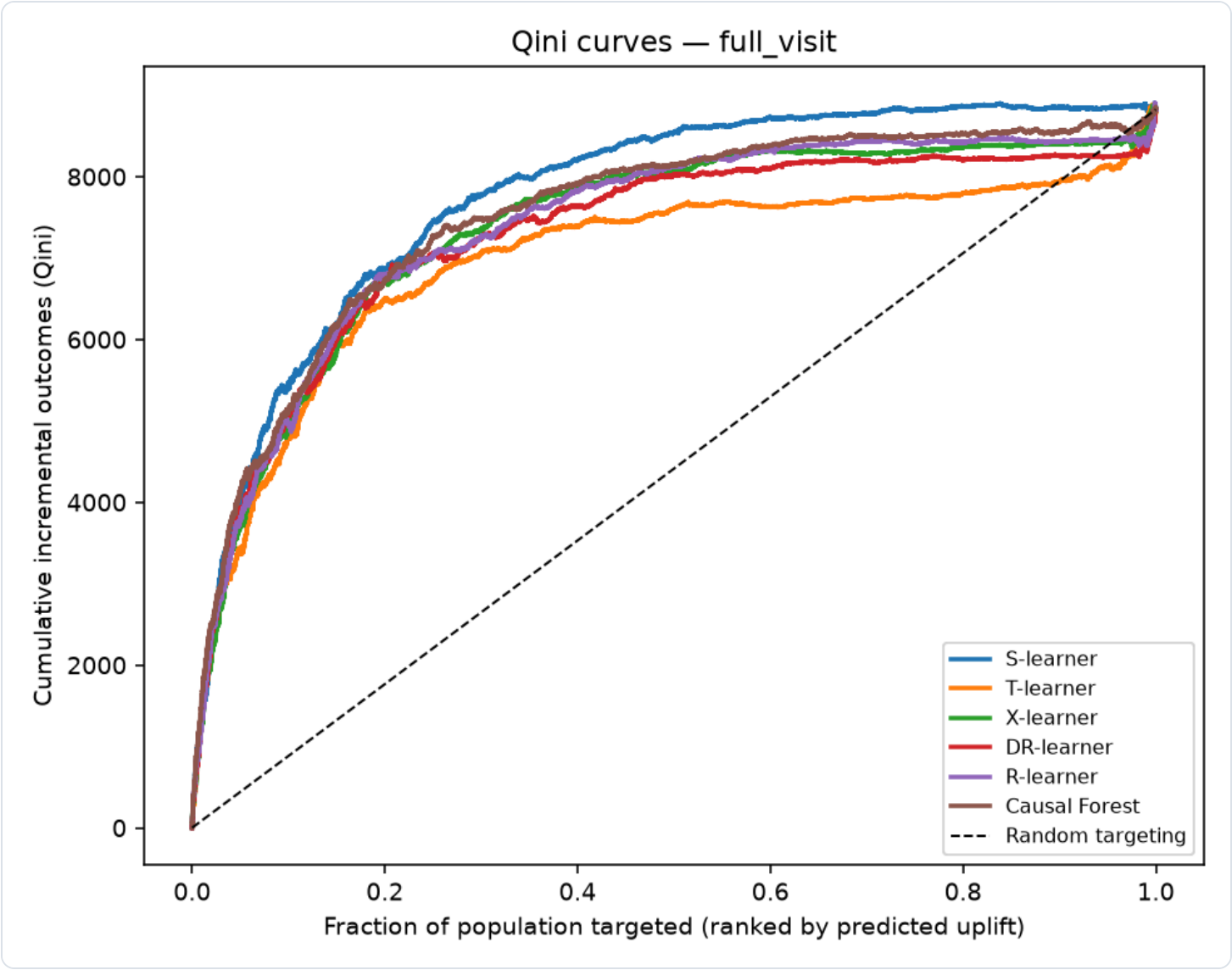


FIG 6 Qini curves, full data

The curves that fanned out at 500k now bunch together. Concretely, the Qini coefficients move like this:

ESTIMATOR	QINI @ 500K	QINI @ 4M	% JUDGED "HARMED" @ 500K → @ 4M
S-learner	0.539	**0.758**	4% → 1%
Causal Forest	0.561	0.701	28% → 23%
R-learner	0.315	0.675	44% → **3%**
X-learner	0.417	0.668	15% → 3%
DR-learner	0.215	0.646	44% → **2%**
T-learner	0.351	0.578	35% → 15%

Three things happen at once:

1. **The spread collapses.** The Qini range narrows from 2.6× (0.22–0.56) to 1.3× (0.58–0.76). With enough data the estimators largely agree on *who* to target — pairwise rank correlation rises from 0.38–0.69 to **0.47–0.89**, and the four cross-fitting methods (S-, X-, DR-, R-learner) now agree at 0.78–0.89.
2. **The data-hungry methods recover.** The DR- and R-learners — dead last at 500k, and the ones the placebo test flagged — climb to the middle of the pack. Their "44% of users are harmed" claim, which looked alarming in §5, was almost entirely finite-sample noise: at 4M it falls to 2–3%. Doubly-robust and DML estimators have more nuisance functions to learn, so they *need* volume; give it to them and they behave.
3. **The ranking reshuffles, but the robust methods stay robust.** The **S-learner** — cheapest to fit — is now the single best ranker (Qini 0.76), with the causal forest a close second. These two were the top pair at 500k too. The lesson is not "S-learner wins"; it is that the *regularized* estimators were trustworthy at both scales, while the flexible ones were only trustworthy at scale.

We can watch all of this happen by sweeping the training size from 50k to 4M and scoring every estimator on one fixed 1M-row evaluation set:

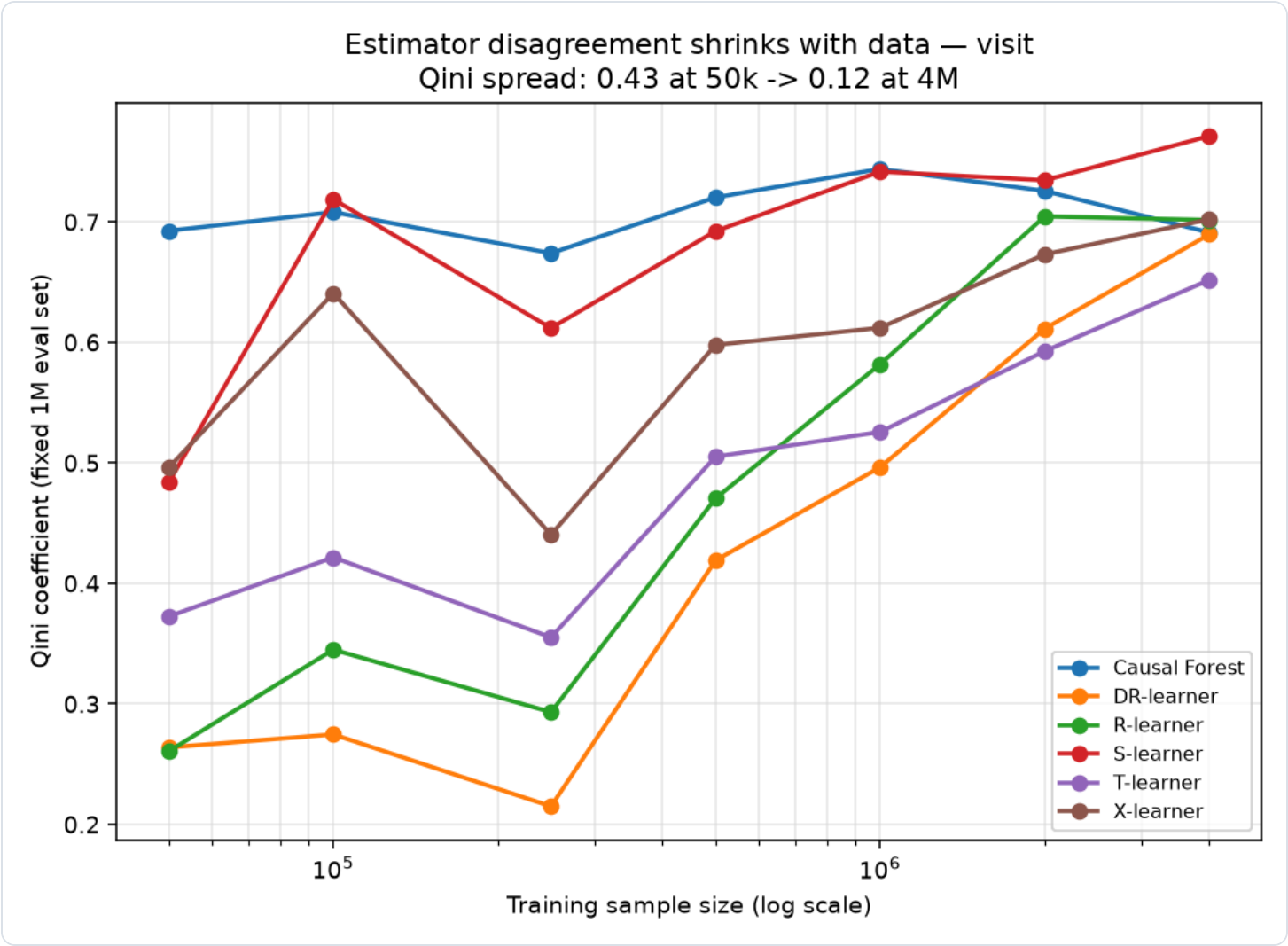


FIG 7 Qini vs training-sample size

The picture is unambiguous. **The causal forest and S-learner are essentially flat and high from 50k onward** — they are the sample-efficient methods, extract most of the ranking signal from a small sample, and barely improve with more. The **DR-, R-, X-, and T-learners start far below (Qini ≈ 0.26–0.37 at 50k) and climb steeply**, only catching up around 2–4M rows. The vertical gap between the best and worst estimator — the cost of picking wrong — shrinks from **0.43 at 50k to 0.12 at 4M.** Estimator choice is a small-sample problem.

Re-running the placebo test at scale closes the loop. In §6 the DR-learner failed its negative control on 500k rows — its real Qini (0.158) sat *below* its own placebo 95th percentile, a z of just 1.4. Repeating that test on the 4M-row training set flips the verdict cleanly:

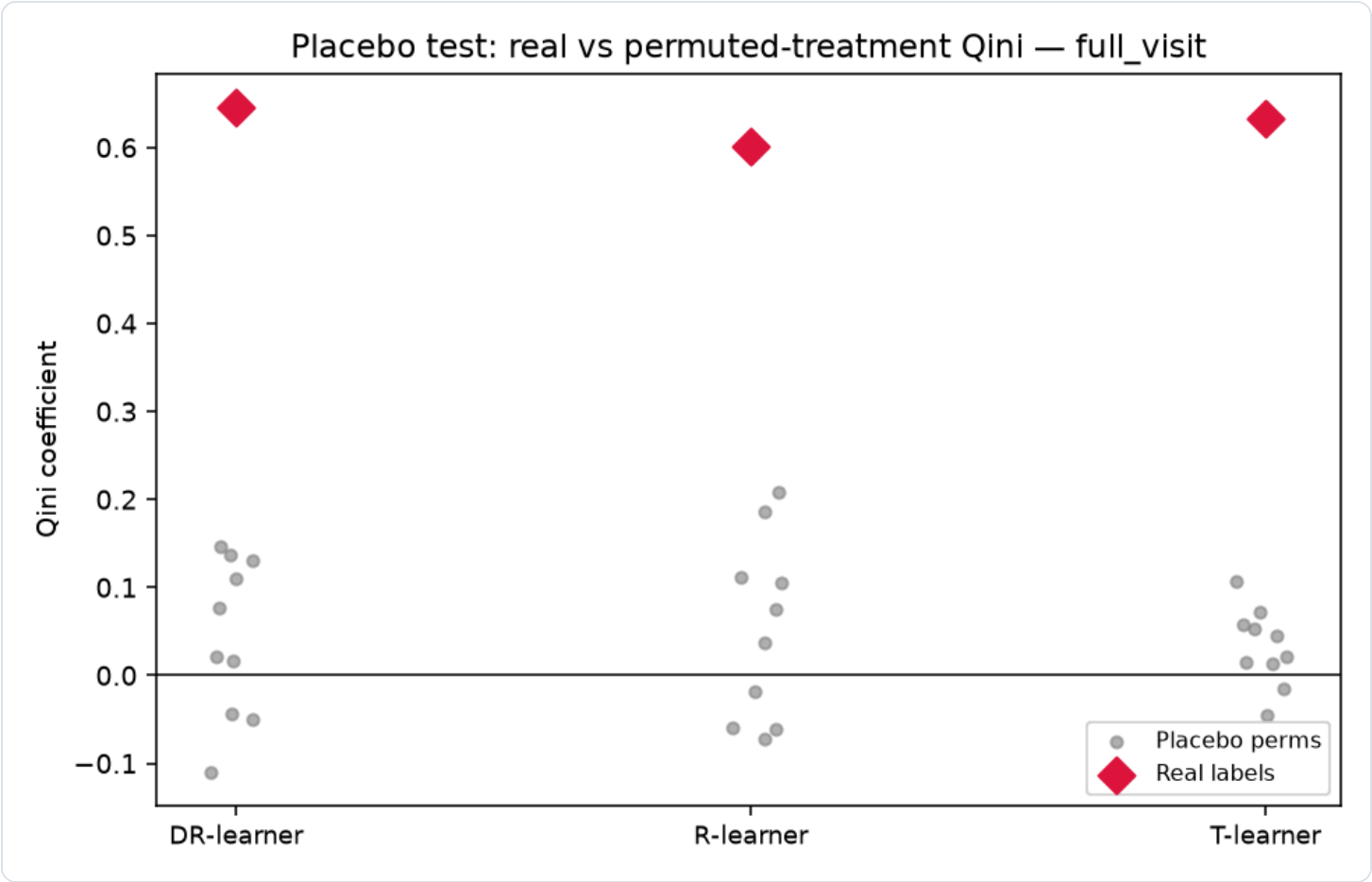


FIG 8 Placebo test on full data

ESTIMATOR	Z VS. NULL @ 500K	Z VS. NULL @ 4M
DR-learner	1.4 (fails)	**7.0**
R-learner	–	5.6
T-learner	3.1	14.4

At 4M the DR-learner clears its permuted-label null by **seven standard deviations** (real Qini 0.646 vs. a null centred at 0.044). Its heterogeneity was real all along — it just needed the data to resolve it above the noise floor. The null tightens with scale too (its spread shrinks as n grows), so more data both strengthens the real signal and sharpens the yardstick you measure it against. This is the cleanest possible confirmation of the section's thesis: the estimator that looked like it was overfitting at 500k was, in fact, *under-powered* — and the placebo test, run at your actual data budget, is what tells the two apart.

And the targeting story gets sharper, not weaker: at 4M the best ranker puts **82% of the incremental visits in the top 10%** of users (95% in the top 30%).

The decision-relevant takeaway: estimator choice matters *most exactly when you have least data* — which is precisely when a team can least afford to guess. With 500k rows the choice swung Qini 2.6×; with 4M it barely

mattered. So the practical rule is: at small scale, prefer regularized estimators (S-learner, causal forest) **and run the placebo check**; as data grows, the DML family becomes safe and the choice matters less.

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8. What a decision-maker should take away

Three things, in the order a decision-maker should care about them.

1. **Targeting is where the money is.** The ATE says "ads work." The Qini curve says *you do not need to treat everyone*: at scale the top 10% of users by predicted uplift carry ~82% of the incremental visits (top 30% → 95%). For any campaign with non-trivial cost-per-contact — a re-engagement push, a credit-line offer — that is the entire business case, and it is invisible in the ATE.
2. **Estimator choice matters most when data is scarce.** At 500k rows the six methods spanned a 2.6× Qini range and disagreed on which users were even *helped* vs *harmed* (4%–44%); at 4M that spread collapsed to 1.3× and the disagreement to 1–3%. Much of the "which estimator?" anxiety is a finite-sample phenomenon. The regularized estimators (S-learner, causal forest) were trustworthy at *both* scales; the doubly-robust / DML methods needed volume. So: small data → prefer regularized methods and verify; large data → the choice is second-order.
3. **The placebo test is the discipline that tells you which regime you're in.** It is cheap (you already have the estimator code), only *available* because the data is randomized, and it is exactly what flagged the DR-learner's 500k heterogeneity as noise — a diagnosis §7 then confirmed by scaling up. Most write-ups of this dataset stop at a single Qini number; benchmarking every estimator against its own permuted-label null, at your actual data budget, is the step that turns "look at my forest" into a defensible recommendation.

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9. Appendix — the rare-event stress test: `conversion`

Everything above used `visit` (base rate ≈ 4.7%). The commercially decisive outcome is `conversion`, but it fires for only **0.29%** of users — roughly 16× rarer. Does the story survive when positive events are scarce? We repeat the full 4M-row run on `conversion`.

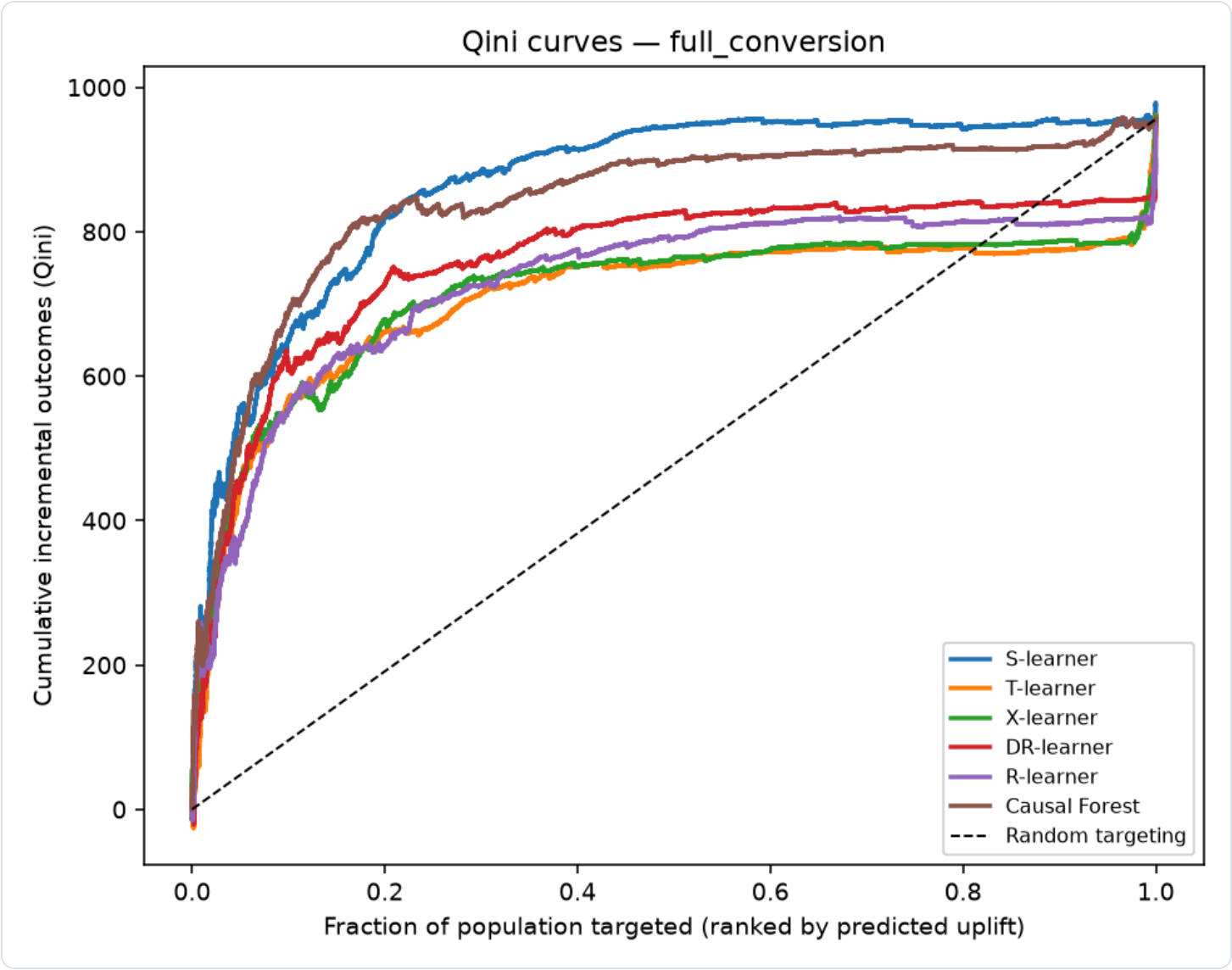


FIG 9 Qini curves, conversion

ESTIMATOR	QINI (CONVERSION)	QINI (VISIT)
S-learner	**0.810**	0.758
Causal Forest	0.757	0.701
DR-learner	0.589	0.646
R-learner	0.523	0.675
X-learner	0.491	0.668
T-learner	0.474	0.578

Two things carry over and one sharpens the §7 thesis.

- **The same two estimators lead.** S-learner and the causal forest are again the top pair, by a clear margin — more evidence that the regularized methods are the safe default.
- **Targeting is, if anything, more valuable here.** The ATE on conversion is a tiny +0.11pp, yet the best ranker still concentrates **82% of the incremental conversions in the top 10%** of users. For the outcome you actually monetise, the ability to target is the whole game.
- **Disagreement widens again — exactly as the sample-size story predicts.** The Qini spread on conversion is **1.7×** (0.47–0.81) versus 1.3× on visit at the same 4M rows. A 0.29% outcome gives you only ~1/16th the *effective* sample of a 4.7% outcome, so 4M conversion rows behave like a much smaller visit dataset — and the estimators fan back out, just as §7 would predict. Rare-event uplift is a small-sample regime in disguise, which is exactly when estimator choice, and the placebo check, matter most.

The three diagnostic views from §4–§5, repeated for `conversion`, make the pattern concrete.

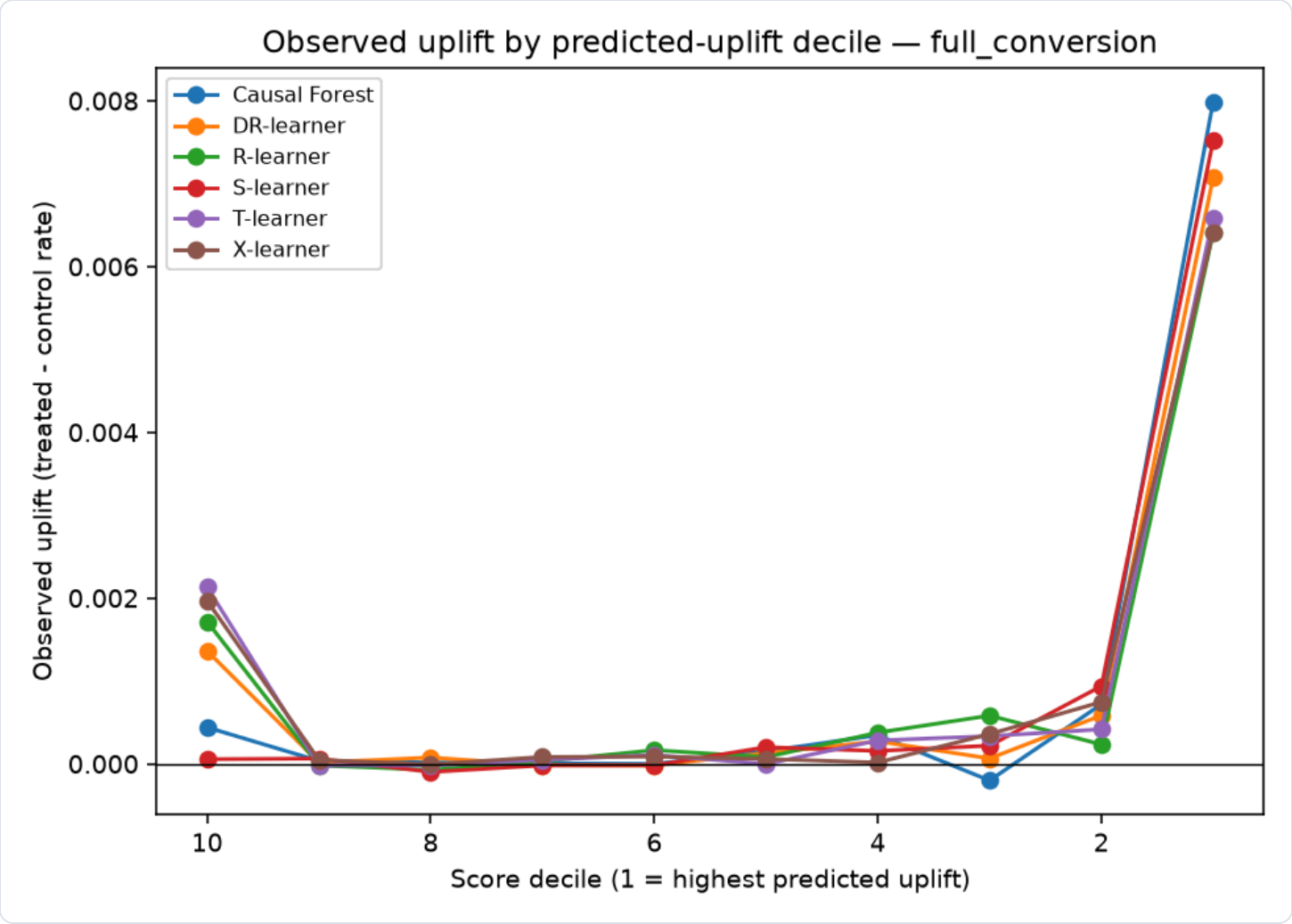


FIG 10 Observed conversion uplift by predicted-uplift decile

The ranking is genuine but *brutally* concentrated: essentially all of the incremental conversions sit in the single top decile (~0.008 observed uplift), with every other decile flat at zero. That is the decile-level face of the "82% in

the top 10%" number — and a reminder that for a 0.29% outcome there is almost no signal to spread across the rest of the population.

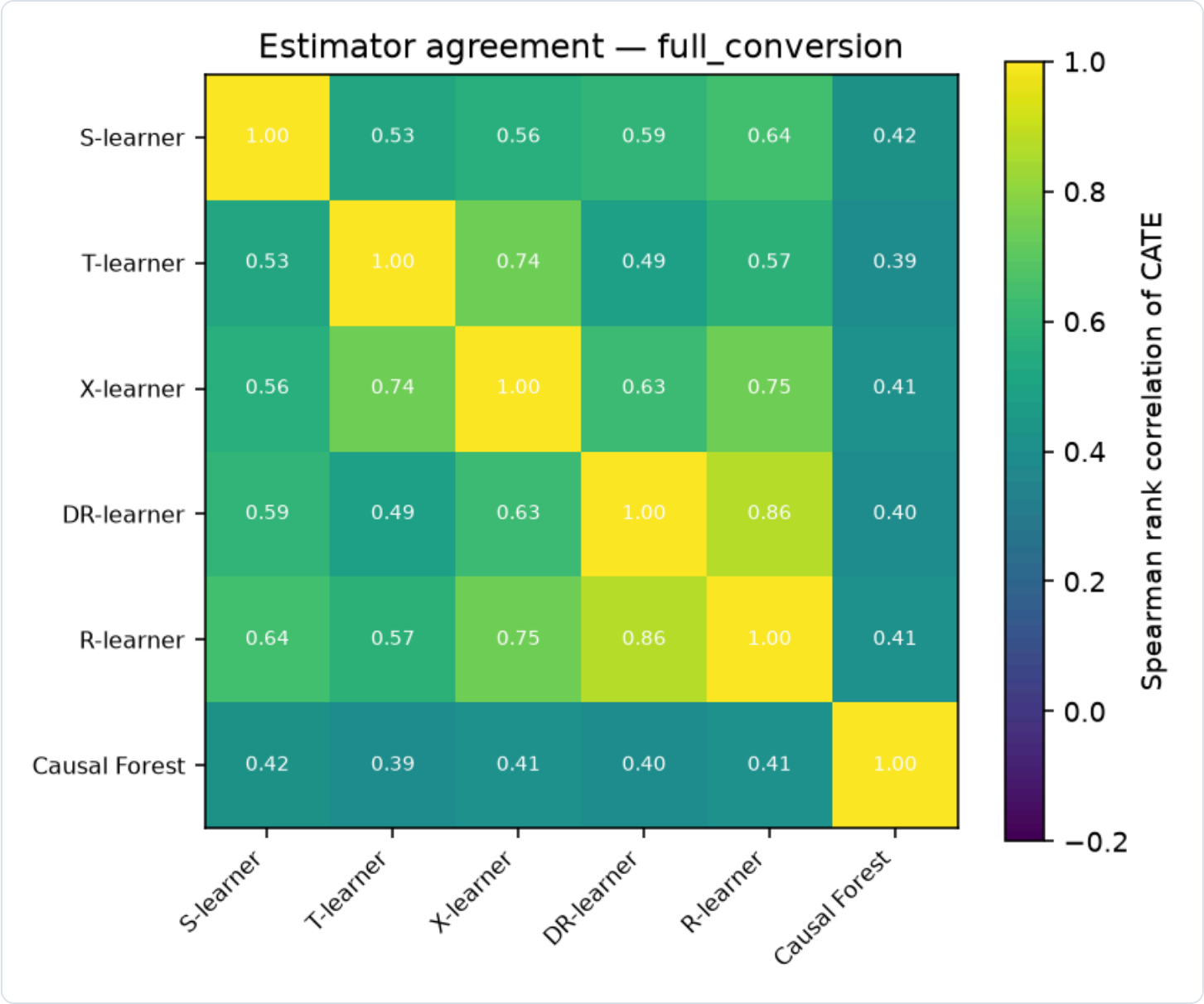


FIG 11 Estimator agreement, conversion

Rank agreement is looser than on `visit` (Spearman 0.39–0.86), and the causal forest is the visible outlier — it correlates only ~0.40 with every other method, because on this rare outcome its honest, deeply-split trees carve out a different (and more negative-tailed) heterogeneity structure than the smoother meta-learners. It still ranks second, so "different" is not "wrong" — but it is a concrete case of two defensible methods disagreeing on *who* to target.

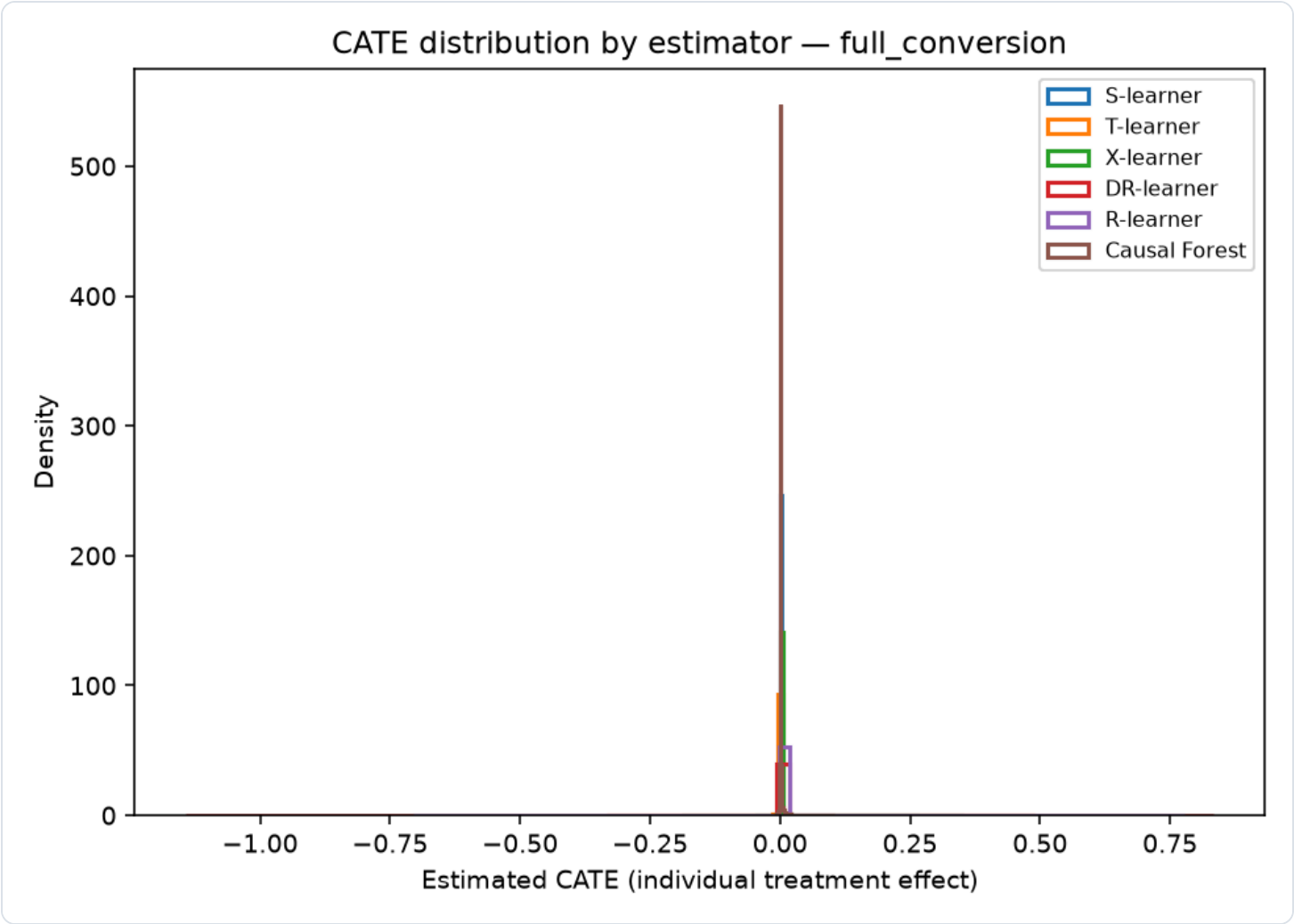


FIG 12 CATE distributions, conversion

The CATE distributions show why the Qini spread widens: the doubly-robust and DML learners (DR, R) carry the heaviest tails at this sample size, while the S-learner and causal forest stay tight around a small positive effect — the same regularized-vs-flexible split seen on `visit`, amplified by the scarcity of conversion events.

10. Reproducibility

All code, the exact environment, and the commands that produced every figure and table are in the repository README. Nothing here depends on a private dataset or a manual step.

Dataset: Diemert, Betlei, Renaudin & Amini, "A Large Scale Benchmark for Uplift Modeling," AdKDD 2018.